Hospital Engineering and Management

Lecture 14: Hospital Water Supply, Plumbing and Sanitation Systems

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Learning Objectives

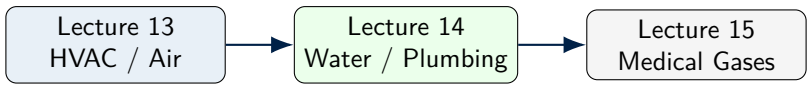
By the end of this lecture, students should be able to:

- explain why water, plumbing and sanitation are patient-safety systems in hospitals;
- identify the complete hospital water pathway from source to point of use and wastewater disposal;
- estimate basic water demand, storage capacity, pumping flow and pipe velocity;
- distinguish potable, non-potable, hot-water, process-water and drainage networks;
- explain water-quality risks, cross-connections, backflow, stagnation and biofilm formation;
- describe hospital sanitation, drainage, sewage-treatment and reuse systems;
- relate water-system engineering to infection prevention, resilience and biomedical equipment safety.

Connection with Lecture 13

Lecture 13 Established

Hospital HVAC manages air as an engineered clinical resource through filtration, ventilation, temperature, humidity and pressure control.



Key Transition

A hospital cannot function safely without reliable water for drinking, hand hygiene, cleaning, sterilization, laboratories, food services, dialysis support, HVAC and sanitation.

Opening Question: What Happens if Water Stops?

Imagine a 500-bed hospital loses its incoming water supply for 12 hours.

- Can toilets and hand-washing stations function?
- Can CSSD process instruments?
- Can the kitchen, laundry and housekeeping continue?
- Can cooling towers and some HVAC systems operate?
- Can laboratories and dialysis services continue normally?
- How long will stored water last?

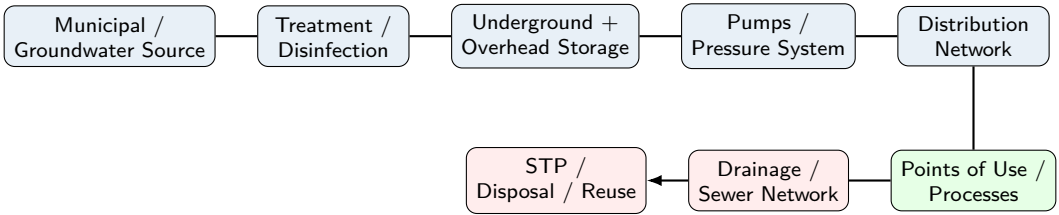
Water failure can become a hospital-wide clinical emergency.

Why Hospital Water Systems Are Different

Compared with a typical commercial building, hospitals have:

- continuous 24/7 occupancy;
- vulnerable and immunocompromised patients;
- high hand-hygiene and cleaning demand;
- specialized process-water requirements;
- critical hot-water demand;
- medical equipment connected to water;
- high consequences of contamination or supply interruption;
- strict separation requirements between clean water and waste systems.

Hospital Water-System Overview



Engineering Principle

Protect water quality from the source all the way to the tap, and keep wastewater physically separated from the potable-water network.

Types of Water Used in Hospitals

- **Potable water:** drinking, hand washing and general domestic use.
- **Hot water:** bathing, cleaning and selected process uses.
- **Treated/process water:** laboratories, CSSD, humidification or equipment-specific use.
- **Dialysis-quality water:** specially treated water for hemodialysis systems.
- **Non-potable/reclaimed water:** selected flushing, landscaping or cooling uses when permitted.
- **Fire-water reserve:** dedicated or protected storage according to fire-system design.

Important

Different water qualities must never be assumed interchangeable. The required quality depends on the clinical process and equipment specification.

Major Sources of Hospital Water

Hospitals may obtain water from:

- municipal/piped public supply;
- groundwater through bore wells where permitted;
- surface-water systems after treatment;
- tanker supply during emergencies;
- rainwater harvesting for approved non-potable uses;
- treated wastewater for approved reuse applications.

Planning Question

Is the source reliable enough for a facility that must operate continuously? If not, storage and alternate supply become part of clinical resilience.

Water Quality: Four Perspectives

Water quality must be considered from four perspectives:

1. **Physical:** colour, turbidity, taste, odour, temperature.
2. **Chemical:** pH, dissolved solids, hardness, chloride, nitrate, metals and disinfectant residual.
3. **Microbiological:** absence/control of indicator organisms and waterborne pathogens.
4. **Process specific:** requirements for dialysis, laboratory analysers, sterilizers and other equipment.

Drinking-Water Quality in India

For potable water, Indian projects commonly refer to **IS 10500:2012, Drinking Water – Specification (Second Revision)**, together with current reviews/amendments and local requirements.

- Water should be acceptable in appearance, taste and odour.
- Microbiological safety is fundamental.
- Chemical parameters have acceptable and, for some parameters, permissible limits.
- Sampling and testing frequency should be risk based and comply with applicable regulations/policies.

Teaching Note

Do not design from a memorized table alone. Always consult the currently applicable BIS document, local authority requirements and the hospital's risk assessment.

Typical Water-Treatment Train

Depending on source-water quality, treatment may include:



Design Principle

Treatment should be selected after water testing; unnecessary treatment can add cost, waste water and create new maintenance risks.

Disinfection Methods

Common approaches include:

- chlorination or other approved chemical disinfection;
- ultraviolet disinfection for selected applications;
- thermal control for parts of hot-water systems;
- point-of-use controls in selected high-risk clinical situations.

Key Point

Disinfection is not a substitute for good system design. Dead legs, stagnation, low flow, scale and biofilm can undermine water quality.

Water Demand: What Creates the Load?

Hospital water demand depends on:

- number of beds and occupancy;
- OPD and visitor load;
- staff strength;
- kitchen and dietary services;
- laundry and linen processing;
- CSSD and sterilization processes;
- laboratories and imaging support;
- cooling towers and HVAC make-up water;
- housekeeping and sanitation;
- landscape and other non-clinical uses.

Basic Water-Demand Model

A simplified planning model is:

$$Q_{daily} = \sum_{i=1}^{n} N_i q_i + Q_{process} + Q_{loss}$$

where:

- N_i = number of users/units in category i;
- q_i = design demand per user/unit;
- $Q_{process}$ = process-water demand;
- Q_{loss} = allowance for leakage, flushing and operational losses.

Professional Practice

The unit-demand values must come from the applicable code, local authority or validated hospital data.

Numerical Example 1: Daily Water Demand

For teaching, assume a hospital has 300 occupied beds and an adopted planning allowance of 450 L/bed/day for the combined hospital demand.

$$Q_{daily} = 300 \times 450 = 135{,}000 \text{ L/day}$$

$$Q_{daily} = 135 \text{ m}^3/\text{day}$$

If a 15% design margin is added:

$$Q_{design} = 135 \times 1.15 = 155.25 \text{ m}^3/\text{day}$$

Teaching Assumption

The 450 L/bed/day value is only an example. Use the applicable project design criterion in real practice.

Water Storage Strategy

Hospitals usually require storage for:

- balancing variable demand;
- interruption of municipal supply;
- emergency resilience;
- fire-fighting requirements;
- operational reserve during maintenance.

Storage Arrangement

A common architecture uses an underground tank/sump, pumping system and elevated or pressure-controlled distribution. Fire-water storage may be segregated or protected from normal consumption.

Numerical Example 2: Storage Capacity

Suppose the design daily demand is 160 m^3/day and the hospital wants 24 hours of usable domestic-water reserve.

$$V_{usable} = 160 \text{ m}^3$$

If 10% additional operational volume is adopted:

$$V_{tank} = 1.10 \times 160 = 176 \text{ m}^3$$

Design Reminder

Actual storage policy should consider source reliability, water age, fire reserve, cleaning downtime and local regulations. Excessive storage can increase stagnation risk.

Hospital Storage Tanks: Engineering Requirements

Good practice includes:

- secure covers and screened vents;
- protection from surface contamination;
- access for inspection and cleaning;
- drainage arrangement for maintenance;
- level monitoring and high/low alarms;
- separate compartments where continuity during cleaning is required;
- overflow routed safely and visibly;
- prevention of cross-connections with sewer or storm-water systems.

Pumping Systems

Hospital water pumping may use:

- transfer pumps from underground to overhead storage;
- hydro-pneumatic/booster systems;
- hot-water circulation pumps;
- sewage and sump pumps;
- STP pumps;
- dedicated process-water pumps.

Reliability

Critical pumps should be selected with appropriate standby capacity, isolation valves, alarms and emergency power where continuity is essential.

Pump Duty Point

A pump must provide the required flow against the total dynamic head:

$$H_{TDH} = H_{static} + H_{friction} + H_{minor} + H_{required\ pressure}$$

- Static head comes from elevation difference.
- Friction head comes from pipe resistance.
- Minor losses occur at valves, bends and fittings.
- Residual pressure must remain at the point of use.

Numerical Example 3: Pump Flow

A 160 m^3/day demand must be transferred to storage in 8 hours of pump operation per day.

$$Q = \frac{160}{8} = 20 \text{ m}^3/\text{h}$$

With a 20% flow margin:

$$Q_{design} = 24 \text{ m}^3/\text{h}$$

If the calculated total dynamic head is 32 m, the preliminary duty point is:

$$\boxed{24 \text{ m}^3/\text{h at } 32 \text{ m head}}$$

Cold-Water Distribution Network

The distribution network includes:

- mains and risers;
- floor/zone isolation valves;
- branches to clinical and support areas;
- pressure-reducing valves where necessary;
- backflow-prevention devices;
- meters and monitoring points;
- flushing points and drain valves;
- appropriate pipe materials and supports.

Pipe Sizing: Basic Relation

The basic flow relation is:

$$Q = Av$$

For a circular pipe:

$$A = \frac{\pi D^2}{4}$$

Therefore:

$$D = \sqrt{\frac{4Q}{\pi v}}$$

Design Trade-off

Too small a pipe causes high velocity, noise and pressure loss; too large a pipe can increase cost and water age/stagnation.

Numerical Example 4: Pipe Diameter

A branch carries 3 L/s. Assume a preliminary design velocity of 1.5 m/s.

$$Q = 0.003 \text{ m}^3/\text{s}$$

$$D = \sqrt{\frac{4(0.003)}{\pi(1.5)}} = 0.0505 \text{ m}$$

$$D \approx 50.5 \text{ mm}$$

A standard pipe size would then be selected and checked for actual velocity and pressure loss.

Pressure Zoning in Multi-Storey Hospitals

Tall hospitals may need multiple pressure zones because:

- lower floors may experience excessive static pressure;
- upper floors may receive insufficient pressure;
- fixtures and equipment have pressure limits;
- pressure-regulating valves or separate booster zones may be required.

Good pressure control = adequate service + reduced leakage + safer fixtures.

Hot-Water Systems

Hot water is required in many hospital areas for:

- bathing and personal hygiene;
- kitchen and food service;
- selected cleaning processes;
- laundry;
- process equipment where specified.

System elements include heaters/calorifiers, storage, recirculation, mixing controls, insulation and temperature monitoring.

Hot-Water Recirculation

Without circulation, distant outlets can have long waiting times and stagnant cooled water.

- hot-water supply runs to fixtures;
- a return line brings water back to the heater;
- circulation pumps maintain movement;
- balancing valves help maintain temperature throughout the loop;
- insulation reduces heat loss.

Safety Challenge

The system must balance microbial-control objectives with prevention of scald injury. Use the facility's water-management plan and applicable guidance.

Numerical Example 5: Heating Energy

Heat 1000 L of water from 20°C to 60°C. Approximate water mass as 1000 kg.

$$Q = mc_p\Delta T$$

$$Q = 1000 \times 4.186 \times 40 = 167{,}440 \text{ kJ}$$

$$Q \approx 46.5 \text{ kWh thermal}$$

If heater efficiency is 90%:

$$E_{input} \approx \frac{46.5}{0.90} = 51.7 \text{ kWh}$$

Cross-Connections and Backflow

A **cross-connection** is an actual or potential connection between potable water and a contaminating source.

- hose submerged in a contaminated sink;
- direct connection to chemical dosing equipment;
- improper connection to process equipment;
- potable and reclaimed-water networks inadvertently connected.

Backflow Prevention

Use appropriate air gaps, vacuum breakers, check devices or reduced-pressure devices according to risk and applicable plumbing requirements.

Back-Siphonage vs Back-Pressure

Back-Siphonage

Potable-system pressure falls below downstream pressure, drawing contaminated liquid backwards.

Back-Pressure

Downstream equipment or process pressure exceeds potable-line pressure and pushes fluid backwards.

Clinical Consequence

A plumbing error can contaminate many outlets before the problem is noticed.

Stagnation and Dead Legs

Stagnant water can occur in:

- unused branches;
- oversized piping;
- vacant patient rooms;
- rarely used taps or showers;
- capped extensions left after renovation;
- poorly balanced hot-water loops.

Engineering Response

Minimize dead legs, maintain intended flow, flush after shutdowns and include low-use outlets in the water-management plan.

Biofilm in Hospital Water Systems

Biofilm is a microbial community attached to a wetted surface and embedded in a protective matrix.

- can develop in pipes, tanks, aerators and shower heads;
- is promoted by stagnation, scale and favourable temperatures;
- can shelter organisms from disinfectants;
- may act as a reservoir for opportunistic pathogens.

Water quality at entry does not guarantee water quality at the outlet.

Legionella and Healthcare Water

Legionella control is especially important in healthcare because susceptible patients may be exposed to aerosols from water systems.

- showers and taps;
- hot-water systems;
- cooling towers and other aerosol-generating systems;
- some medical-water devices if poorly managed.

Primary Control Strategy

Use a multidisciplinary water-management program that identifies hazardous conditions, control measures, monitoring and corrective actions.

Water Management Program: Core Elements

A hospital water-management program should conceptually include:

1. multidisciplinary team;
2. system description and process-flow diagram;
3. identification of high-risk areas and hazardous conditions;
4. control measures and limits;
5. monitoring procedures;
6. corrective actions;
7. verification and validation;
8. documentation and review.

High-Risk Water Locations in Hospitals

Special attention may be required for:

- ICUs and transplant/protective units;
- dialysis services;
- neonatal and oncology areas;
- ice machines and drinking-water dispensers;
- sinks near sterile or medication-preparation areas;
- hydrotherapy equipment;
- endoscope reprocessing systems;
- low-use outlets and dead legs.

Sink Design and Splash Risk

Sinks can become reservoirs and can generate contaminated splash or droplets.

- locate sinks to avoid splash onto clean supplies;
- avoid storing sterile items near basins;
- design faucets and basins to reduce direct impingement on drains;
- keep traps and drains maintained;
- manage hand-wash sinks differently from utility/dirty sinks.

Water for Hemodialysis

Hemodialysis water receives special treatment because large volumes indirectly contact the patient's blood through the dialyzer membrane.

- pretreatment may include sediment filtration, softening and carbon filtration;
- reverse osmosis is commonly central to the treatment train;
- distribution-loop design and disinfection are critical;
- chemical and microbiological quality must meet dialysis-specific standards;
- routine monitoring and documentation are essential.

Important

Potable water is only the feed water; it is not automatically suitable as dialysis water.

Water for CSSD and Sterilizers

Poor water quality can cause:

- scale in boilers and steam generators;
- corrosion or staining of instruments;
- poor washer-disinfector performance;
- residue on devices;
- increased maintenance and energy use.

Engineering Coordination

CSSD, biomedical engineering and facility engineering should verify equipment-manufacturer requirements for feed water, final rinse water and steam quality.

Laboratory and Analyzer Water

Many laboratory systems require purified water beyond normal potable-water quality.

- reagent preparation;
- chemistry analyzers;
- glassware rinsing;
- molecular or high-purity applications.

Treatment may include RO, deionization, ultrafiltration or polishing depending on equipment specification.

Plumbing Fixtures in Patient Areas

Patient-care plumbing may include:

- hand-wash basins;
- showers and bathing fixtures;
- toilets and bedpan-cleaning facilities;
- utility sinks;
- clinical scrub sinks;
- drinking-water points.

Fixture selection should consider accessibility, cleanability, infection risk, water efficiency and maintenance.

Sanitation: More Than Toilets

Hospital sanitation includes:

- adequate toilets for patients, staff and visitors;
- accessible toilets for persons with disabilities;
- hand-hygiene facilities;
- safe disposal of excreta;
- drainage of showers and wet areas;
- cleaning and maintenance schedules;
- odour and pest control;
- separation of clean and contaminated activities.

Building Drainage System

The sanitary drainage network carries wastewater from fixtures to the external sewer/STP system.

- soil pipes;
- waste pipes;
- traps;
- vent pipes;
- stacks and branches;
- cleanouts and inspection chambers;
- building sewer connection.

Why Traps Are Essential

A trap maintains a water seal that blocks sewer gases from entering occupied spaces.

- floor traps;
- basin traps;
- WC traps;
- equipment traps where appropriate.

Failure Modes

Trap seals can be lost by evaporation, siphonage, back pressure, leakage or poor design, allowing odour and potentially contaminated air movement.

Vent Pipes and Drainage Ventilation

Drainage ventilation helps:

- equalize pressure in the drainage system;
- protect trap seals;
- allow gases to discharge safely outdoors;
- improve drainage performance.

Design Principle

A drainage system must remove wastewater without allowing foul air, leakage or cross-contamination into clinical spaces.

Hospital Wastewater Characteristics

Hospital wastewater may contain:

- normal domestic sewage;
- detergents and cleaning chemicals;
- pharmaceutical residues;
- laboratory contaminants;
- microorganisms;
- oils/grease from kitchens;
- high-temperature or process discharges.

Some specialty wastes require pretreatment or separate management according to applicable regulations and risk assessment.

What Should Not Enter the Normal Drain?

Examples requiring specific management may include:

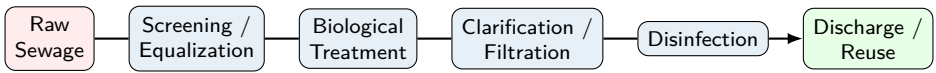
- concentrated chemicals and solvents;
- cytotoxic residues;
- mercury or heavy-metal-containing waste;
- radioactive liquids beyond authorized limits;
- large quantities of pharmaceutical waste;
- infectious or laboratory waste requiring special treatment.

Rule

Never use the drain as a universal waste-disposal route. Follow biomedical, hazardous, chemical and radiation-waste procedures.

Sewage Treatment Plant (STP): Purpose

An STP reduces pollutants before discharge or approved reuse.



Common Biological Treatment Approaches

Hospital STPs may use processes such as:

- activated sludge;
- moving-bed biofilm reactor (MBBR);
- sequencing batch reactor (SBR);
- membrane bioreactor (MBR);
- other approved biological treatment systems.

Selection depends on flow, footprint, influent characteristics, effluent targets, operator skill, energy and lifecycle cost.

Numerical Example 6: Preliminary STP Capacity

Assume daily water consumption is 160 m^3/day and, for a teaching estimate, 80% becomes sewage.

$$Q_{sewage} = 0.80 \times 160 = 128 \text{ m}^3/\text{day}$$

If a 20% capacity margin is used:

$$Q_{STP} = 1.20 \times 128 = 153.6 \text{ m}^3/\text{day}$$

A preliminary nominal capacity might therefore be about 155–160 m^3/day before detailed process design.

Greywater and Reuse

Treated water may be reused for approved non-potable purposes such as:

- toilet flushing;
- landscaping;
- cooling-tower make-up where technically suitable;
- selected cleaning uses where permitted.

Critical Requirement

Reuse networks require clear identification, physical separation and backflow protection to prevent accidental connection with potable water.

Dual-Piping Systems

A dual-piping system separately distributes:

- potable water; and
- treated/reclaimed non-potable water.

Engineering controls should include:

- distinct identification and labelling;
- separate valves and meters;
- cross-connection testing;
- controlled commissioning and documentation;
- staff awareness.

Storm-Water Drainage

Storm water should be managed separately from sanitary sewage wherever required by the local system.

- roof drains and rainwater pipes;
- surface drains;
- site grading;
- recharge/harvesting systems;
- flood protection for basements and critical equipment.

Hospital Risk

Flooding can disable electrical rooms, oxygen manifolds, pumps, data centres and access routes even when patient floors remain dry.

Basement and Sump Pumping

Below-grade drainage may require sump pumps.

- duplex duty/standby pumps;
- automatic level control;
- high-level alarm;
- non-return valves;
- emergency power where failure could cause flooding;
- maintenance access.

Kitchen and Food-Service Drainage

Kitchen plumbing should address:

- potable-water hygiene;
- hot-water demand;
- grease interception where required;
- floor drainage and cleanability;
- prevention of pest access;
- separation of food preparation from waste handling.

Laundry Water and Drainage

Laundry systems may have high requirements for:

- water volume;
- temperature;
- hardness control;
- detergent/chemical handling;
- floor drainage;
- wastewater temperature and load.

Water treatment and heat-recovery opportunities should be evaluated against infection-control and equipment requirements.

Cooling-Tower Water

Cooling towers are part of many central HVAC plants and require water-management attention.

- evaporation concentrates dissolved solids;
- blowdown controls cycles of concentration;
- chemical treatment reduces scale, corrosion and biological growth;
- drift can generate aerosols;
- maintenance and monitoring are essential.

Water Metering and Sub-Metering

Metering helps answer:

- How much water does the hospital consume?
- Which departments are the largest users?
- Is there a hidden leak?
- Did a conservation project actually save water?
- Is the STP receiving expected flow?

Useful sub-meter locations include laundry, kitchen, cooling towers, dialysis, CSSD and major buildings/wings.

Leak Detection

Signs of leakage include:

- unexplained night-time flow;
- falling tank level without demand;
- damp walls or ceilings;
- repeated pump cycling;
- abnormal water bills;
- pressure loss;
- acoustic or thermal indications.

BMS Opportunity

Trend tank level, pump status, pressure and flow together to detect abnormal consumption patterns.

Water Conservation in Hospitals

Possible measures include:

- low-flow fixtures where clinically appropriate;
- sensor/controlled taps with proper infection-risk assessment;
- leak-reduction programs;
- efficient laundry equipment;
- cooling-tower optimization;
- rainwater harvesting;
- treated-water reuse;
- sub-metering and benchmarking.

Priority

Water efficiency must not reduce hand hygiene, cleaning effectiveness, equipment performance or infection prevention.

Water Balance for a Hospital

A simple water balance is:

$$\text{Total Inflow} = \text{Measured Uses} + \text{Wastewater} + \text{Evaporation} + \text{Losses}$$

A water balance can reveal:

- unmetered use;
- leakage;
- excessive cooling-tower blowdown;
- abnormal department consumption;
- opportunities for reuse.

Water-System Instrumentation

Useful monitoring points include:

- tank level transmitters;
- flow meters;
- pressure sensors;
- pump current and status;
- conductivity/TDS where relevant;
- disinfectant residual where required;
- temperature in hot-water loops;
- STP pH, dissolved oxygen and process parameters;
- leak-detection sensors in critical rooms.

BMS and Alarm Integration

Critical water alarms can include:

- low domestic-water tank level;
- high tank level/overflow;
- pump trip;
- low distribution pressure;
- abnormal high flow;
- hot-water temperature deviation;
- STP process alarm;
- basement sump high-high level.

Preventive Maintenance: Water Supply

A preventive-maintenance plan should address:

- pumps, motors and bearings;
- tank inspection and cleaning;
- valves and pressure-reducing stations;
- backflow-prevention devices;
- hot-water heaters and circulation pumps;
- water-treatment equipment;
- meters and sensors;
- pipe supports, corrosion and leakage.

Preventive Maintenance: Drainage and STP

Key activities include:

- drain and trap inspection;
- cleaning of grease traps and interceptors;
- sewer-line blockage prevention;
- sump-pump testing;
- STP blower, pump and diffuser maintenance;
- sludge handling;
- effluent testing;
- odour-control inspection.

Emergency Water-Supply Planning

Hospitals should plan for:

- municipal supply interruption;
- source contamination advisory;
- pump or power failure;
- tank contamination;
- major pipe burst;
- flood or natural disaster;
- tanker or alternate-source connection.

Priority Planning

Identify which clinical services must receive water first and which activities can be reduced or suspended temporarily.

Emergency Water Prioritization

During a severe shortage, consider priority for:

1. drinking and essential patient care;
2. hand hygiene and infection control;
3. toilets and sanitation;
4. critical sterilization / reprocessing;
5. kitchen and essential food preparation;
6. critical equipment/processes;
7. other support and nonessential uses.

Actual prioritization must be defined in the hospital emergency plan.

Pipe Burst in a Clinical Area

Immediate actions may include:

- isolate the affected zone;
- protect patients and electrical/medical equipment;
- stop water migration to lower floors;
- coordinate infection-control assessment;
- repair and disinfect/flush as required;
- verify water quality before returning affected outlets to service;
- investigate root cause.

Water Intrusion and Mold Risk

A plumbing leak can become an infection-control problem when building materials remain wet.

- remove standing water quickly;
- identify contaminated porous materials;
- dry the area;
- protect high-risk patients from construction/remediation dust;
- verify hidden spaces above ceilings and behind walls;
- coordinate engineering and infection-control teams.

FMEA Example: Domestic-Water Booster System

Failure	Effect	Detection	Control
Duty pump trip	low pressure / no water	pressure alarm	auto standby pump
Tank low level	supply interruption	level alarm	reserve + alternate source
PRV failure	excessive pressure	pressure trend	relief / maintenance
Check valve failure	backflow / cycling	flow-pressure pattern	inspection / redundancy

FMEA Example: Hospital Drainage

Failure	Effect	Detection	Control
Blocked stack	overflow / service loss	slow drainage	cleanouts + PM
Dry trap	sewer gas entry	odour / inspection	trap seal maintenance
Sump pump trip	basement flooding	high-level alarm	duty-standby + backup power
Cross-connection	water contamination	testing / incident	prevention + labelling

Water and Sanitation Performance Indicators

Possible KPIs include:

- water consumption per occupied bed-day;
- percentage of potable-water tests meeting requirements;
- unplanned water-supply interruptions;
- number of significant leaks per month;
- pump PM compliance;
- percentage of STP effluent samples compliant;
- proportion of water use from approved reclaimed sources;
- time to restore critical plumbing failures.

Role of the Biomedical Engineer

Biomedical engineers should:

- identify medical devices that depend on water supply or drainage;
- communicate feed-water quality, pressure and temperature requirements;
- coordinate installation of dialysis, CSSD, laboratory and endoscopy equipment;
- investigate equipment failures linked to scale, pressure, temperature or water quality;
- ensure device drains do not create unsafe cross-connections;
- participate in commissioning and risk assessment.

Role of the Hospital / Facility Engineer

Facility engineers are responsible for:

- reliable water supply and pressure;
- tank, pump and pipe maintenance;
- hot-water and treatment systems;
- plumbing and drainage integrity;
- STP operation and monitoring;
- water-metering and conservation;
- emergency response;
- compliance documentation and coordination with infection control.

Multidisciplinary Water Governance

Safe hospital water requires collaboration among:

- facility engineering;
- infection prevention and control;
- biomedical/clinical engineering;
- microbiology and laboratory services;
- nursing and clinicians;
- housekeeping and CSSD;
- dialysis specialists where applicable;
- quality and hospital administration.

Water safety is both an engineering and clinical responsibility.

Current Standards and Guidance

For design and operation, consult the requirements adopted by the authority having jurisdiction.

- **National Building Code of India 2016, Part 9**: Plumbing Services, including water supply and drainage/sanitation.
- **IS 10500:2012**: Drinking Water – Specification (Second Revision), with current review/amendments.
- **WHO Global Framework for Action 2024–2030**: universal WASH, waste and electricity services in health-care facilities.
- **WHO WASH FIT**: risk-based improvement of water, sanitation and hygiene in healthcare facilities.
- **CDC healthcare water / Legionella guidance**: water-management programs and environmental infection control.

Case Study 1: Water Supply Stops at 2 AM

A 350-bed hospital receives an alarm that the underground tank is at 15% and municipal supply has stopped.

1. Verify the level and source failure.
2. Estimate remaining hours of water at current demand.
3. Activate alternate supply arrangements.
4. Inform hospital incident command/administration.
5. Reduce nonessential consumption.
6. Protect critical clinical and sanitation functions.

Engineering Lesson

Tank level is a clinical-resilience parameter, not merely a plumbing parameter.

Case Study 2: Repeated Pseudomonas in an ICU

Environmental investigation suggests several ICU sink drains may be reservoirs.

- review sink and faucet geometry;
- inspect splash onto nearby surfaces;
- assess cleaning and drain practices;
- review water temperatures and stagnation;
- coordinate microbiology, infection control and engineering;
- avoid unplanned plumbing changes without risk assessment.

Case Study 3: Dialysis RO Alarm

The dialysis RO plant alarms for conductivity during treatment hours.

- follow dialysis clinical safety procedure immediately;
- verify instrument reading and treatment-train status;
- check pretreatment, RO membranes and distribution loop;
- communicate with dialysis staff and biomedical engineering;
- document corrective action and water-quality verification.

Lesson

A facility-water problem can become a direct medical-device and patient-safety problem.

Case Study 4: Basement Flooding

A storm causes water to enter the basement where electrical panels and water pumps are located.

- protect life and electrical safety first;
- isolate affected power circuits if required;
- activate sump/dewatering systems;
- protect pumps, controls and emergency systems;
- inspect for contamination before restart;
- review site drainage and flood barriers after the event.

Case Study 5: Hot-Water Complaint

A distant ward reports that hot water takes 8 minutes to arrive and becomes lukewarm quickly.

- check recirculation-pump operation;
- measure supply and return temperatures;
- inspect balancing valves;
- verify pipe insulation;
- check for dead legs or recent modifications;
- include infection-control implications in the investigation.

Classroom Activity 1: Tank Sizing

A hospital's design water demand is 210 m^3/day.

1. Estimate 18 hours of usable domestic-water storage.
2. Add a 10% operational allowance.
3. Propose whether the tank should be single or divided into compartments.
4. List three alarms/monitoring points.

Work in groups for 6 minutes.

Classroom Activity 1: Quick Solution

For 18 hours:

$$V = 210 \times \frac{18}{24} = 157.5 \text{ m}^3$$

With 10% allowance:

$$V_{design} = 157.5 \times 1.10 = 173.25 \text{ m}^3$$

A preliminary selection might be about 175 m^3 usable capacity, subject to code, fire reserve and operational strategy.

- low-level alarm;
- high-level/overflow alarm;
- inlet/source flow or pump-status monitoring.

Classroom Activity 2: Find the Hidden Risk

Identify the main engineering risk in each scenario:

1. A hose is left submerged in a dirty utility sink.
2. A rarely used clinic tap has not been opened for six months.
3. Reclaimed-water pipework is the same colour as potable pipework.
4. A basement sump has only one pump and no alarm.
5. A new analyzer is connected without checking required water quality.

Mini-Project Ideas

- Hospital water-balance and leakage audit.
- IoT tank-level and pump-health monitoring system.
- Low-cost smart water-meter dashboard for hospital departments.
- Risk map for stagnation and dead legs in a hospital wing.
- Dialysis-water treatment monitoring architecture.
- STP performance dashboard with alarm logic.
- Rainwater/reclaimed-water feasibility study for a hospital campus.

Summary

- Hospital water systems support clinical care, hygiene, processes and sanitation.
- Source quality, treatment, storage, pumping and distribution must be managed as one system.
- Cross-connections, stagnation, biofilm and pressure failures can create patient-safety risks.
- Dialysis, CSSD and laboratories may require specialized water quality.
- Drainage, STP and reuse systems must protect both public health and the environment.
- Monitoring, maintenance, water-management programs and emergency planning sustain safe operation.
- Biomedical and facility engineers must coordinate whenever medical equipment depends on water.

In a hospital, safe water is an engineered clinical utility.

Assignment

Prepare a two-page engineering note for a **300-bed hospital water and sanitation system**.

1. Draw the flow from source to storage, treatment, distribution and drainage.
2. Identify five major water-demand categories.
3. Propose an emergency-water strategy.
4. Identify three cross-connection or stagnation risks.
5. Show the STP and one possible treated-water reuse route.
6. List five parameters/alarms to monitor through BMS or a dashboard.

Include

At least one numerical estimate: daily demand, storage size, pump flow or STP capacity.

Suggested Reading

- Bureau of Indian Standards, *National Building Code of India 2016*, Part 9: Plumbing Services.
- Bureau of Indian Standards, *IS 10500:2012 – Drinking Water Specification (Second Revision)*, current review/amendments.
- WHO, *Universal Water, Sanitation, Hygiene, Waste and Electricity Services in All Health Care Facilities: Global Framework for Action 2024–2030*.
- WHO, *WASH FIT: A Practical Guide for Improving Quality of Care Through WASH in Health Care Facilities*, 2nd ed.
- CDC, *Environmental Infection Control: Water* and healthcare water-management guidance.
- G. D. Kunders, *Hospitals: Facilities Planning and Management*.

Next Lecture: Centralized Medical Gas and Vacuum Systems

Thank You

Questions and Discussion

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